Business Analytics — Solved Answer Sheet

MBA Semester I • 2024 Pattern • Simple, exam-friendly English

Study note: The uploaded Business Analytics paper has one session. All visible questions and internal OR choices are answered.

Answers include all visible questions and internal OR options in the uploaded paper(s).

May 2025 Paper

Q1(a) Data and information [2 marks]

Data are raw facts such as numbers or names. Information is processed and organised data that is meaningful for decision-making.

Q1(b) Data cleaning [2 marks]

Data cleaning means identifying and correcting or removing errors, duplicates, missing values and inconsistent entries in a dataset.

Q1(c) Two objectives of financial analytics [2 marks]

Two objectives are to forecast revenue/cash flow and to evaluate financial performance, profitability and risk.

Q1(d) Talent management [2 marks]

Talent management is the process of attracting, selecting, developing, motivating and retaining employees with the skills needed by an organisation.

Q1(e) Data visualisation [2 marks]

Data visualisation presents data using charts, graphs, maps or dashboards so that patterns and comparisons are easier to understand.

Q1(f) Two benefits of operational analytics [2 marks]

It identifies process bottlenecks and helps reduce cost, waste and delays while improving productivity.

Q1(g) Predictive analytics [2 marks]

Predictive analytics uses historical data, statistical methods and models to estimate likely future outcomes.

Q1(h) Business analytics tools [2 marks]

Examples include Microsoft Excel, SQL, Power BI, Tableau, Python, R and statistical software.

Q2(a) Importance of business analytics in healthcare [5 marks]

Healthcare analytics helps predict patient risks, improve diagnosis and treatment planning, monitor hospital performance, manage beds and staff, reduce waiting times and control costs. It can also identify disease trends. Patient data must be kept private and secure.

Q2(b) Five Vs of Big Data [5 marks]

Volume means the large amount of data; Velocity means the speed at which data is created and processed; Variety means different formats such as text, images and numbers; Veracity means data quality and reliability; Value means the useful insight gained from the data.

Q2(c) Secondary data collection, privacy and security [5 marks]

Secondary data is data already collected by another person or organisation. Sources include government reports, published research, company records, websites, databases and industry reports. Privacy means using personal data fairly and only for proper purposes. Security means protecting data through access controls, encryption, backups and safe storage. Organisations should obtain data lawfully and avoid sharing sensitive information without permission.

Q3(a) Marketing analytics, segmentation, targeting and positioning [10 marks]

Marketing analytics uses sales, customer and campaign data to measure performance and improve decisions. Segmentation divides a market by geography, demographics, lifestyle or behaviour. Targeting selects the segment that best matches company resources and objectives. Positioning creates a clear image of the product in the customer's mind. Example: a sports shoe company can segment by age and activity, target college athletes, and position the shoe as affordable and durable. Analytics measures which campaigns and segments produce the best results.

Q3(b) Financial forecasting and planning using pro forma statements [10 marks]

Financial forecasting estimates future sales, expenses, profit and cash flow. Financial planning uses these estimates to set budgets, arrange funds and control spending. A pro forma income statement forecasts revenue and expenses; a pro forma balance sheet estimates future assets, liabilities and equity; a cash-flow forecast estimates cash receipts and payments. Steps: study historical results, estimate sales, forecast costs, calculate profit, estimate working capital and cash needs, then compare actual results with the forecast and update assumptions.

Q4(a) Workforce planning, talent management, engagement and performance measurement [10 marks]

HR analytics uses employee data to improve workforce decisions. Workforce planning estimates future staffing needs and skill gaps. Talent management attracts, trains, develops and retains suitable employees. Employee engagement measures commitment, motivation and satisfaction through surveys, feedback and participation. Performance measurement uses clear indicators such as productivity, quality, attendance, goal achievement and customer feedback. Analytics can identify turnover risk and training needs. Data should be used fairly and confidentially.

Q4(b) Operational analytics and process optimisation [10 marks]

Operational analytics examines data from daily business processes to improve efficiency. It measures output, cycle time, defects, downtime, inventory and cost. Process optimisation identifies bottlenecks and removes unnecessary steps. For example, a warehouse can analyse order data to place fast-moving products near packing stations. Steps include collecting reliable data, defining performance indicators, finding the main problem, testing an improvement and monitoring results. Benefits include lower cost, faster delivery and better quality.

Q5(a) Healthcare analytics: patient care and resource management [10 marks]

Healthcare analytics uses patient, treatment and hospital-operation data to improve care. It can help identify high-risk patients, support early diagnosis, personalise treatment, monitor readmissions and improve appointment scheduling. For resource management, it forecasts demand for beds, staff, medicines and equipment. Managers can use dashboards to reduce waiting times and avoid shortages. Example: past admission data can help a hospital plan staffing for busy periods. Patient privacy, data quality and professional judgement must be maintained.

Q5(b) Agribusiness analytics case study [10 marks]

Problem: agribusiness faces changing prices, unpredictable weather and sustainability demands. Data to collect: soil and crop records, rainfall and temperature, seed and fertiliser use, labour, equipment, market prices and sales. Descriptive analytics shows past yield and cost; predictive analytics forecasts weather-related risks, crop yield and prices; prescriptive analytics recommends planting schedules, irrigation and input levels. A dashboard can compare yield per acre, cost per unit and water use. The business can use forecasts to choose crops, reduce waste, plan storage and select selling times. Start with a pilot farm, compare results with previous seasons and expand successful practices. Protect data and review forecasts regularly.